



SPENDWISE - PROJECT PLANNING PHASE

Version: 1.0.0 | Date: March 7, 2026 | Status: ✔ Completed

1. PROJECT OVERVIEW

Attribute	Details
Project Name	Spendwise - Personal Expense Tracker
Version	1.0.0
Repository Path	D:\SPENDWISE
Technology Stack	MERN (MongoDB, Express.js, React, Node.js) - Backend Complete
Lead Developer	Ganesh
Start Date	March 1, 2026
Completion Date	March 7, 2026 (Backend Phase)

2. PROJECT STRUCTURE

```
D:\SPENDWISE\  
├─ server.js # Main entry point (port 5001)  
├─ .env # Environment variables  
├─ package.json # Dependencies  
├─ config/  
│ └─ db.js # MongoDB connection  
├─ models/  
│ └─ User.js # User schema with bcrypt  
│ └─ Transaction.js # Transaction schema  
├─ routes/  
│ └─ authRoutes.js # Authentication endpoints  
│ └─ transactionRoutes.js # Transaction endpoints  
├─ middleware/  
│ └─ authMiddleware.js # JWT verification
```

```
└─ test scripts/
└─ test-spendwise.ps1 # PowerShell test script
└─ test.ps1 # API test script
```

3. DEVELOPMENT TIMELINE

Phase	Duration	Key Deliverables	Status
Planning	Week 1	Project structure, requirements	✔ Complete
Backend Development	Week 2-3	API endpoints, database models	✔ Complete
Testing	Week 4	API testing, UAT	✔ Complete
Deployment	Week 5	Production deployment	⌚ Pending

4. ENVIRONMENT CONFIGURATION

```
PORT=5001
MONGO_URI=mongodb://127.0.0.1:27017/spendwise
JWT_SECRET=spendwise_secret_key
```

5. DEPENDENCIES

```
{ "dependencies": { "express": "^4.18.2", "mongoose": "^7.5.0",
"dotenv": "^16.3.1", "bcryptjs": "^2.4.3", "jsonwebtoken":
"^9.0.2", "cors": "^2.8.5" } }
```

6. RISK MANAGEMENT

Risk	Mitigation	Status
Database connection issues	Use 127.0.0.1 instead of localhost	✔ Resolved
JWT security	30-day expiry, secure secret key	✔ Implemented
Port conflicts	Configurable PORT in .env (5001)	✔ Resolved
Database name conflict	Using lowercase 'spendwise'	✔ Resolved

7. PROJECT MILESTONES ACHIEVED

- ☒ Server running on port 5001
- ☒ MongoDB connected successfully
- ☒ User registration with bcrypt hashing
- ☒ JWT authentication working
- ☒ CRUD operations for transactions
- ☒ Statistics API working
- ☒ Test scripts passing all tests

8. NEXT PHASES

1. **Phase 2:** Problem Definition - Complete
2. **Phase 3:** Solution Requirements - Complete
3. **Phase 4:** Technology Stack - Complete
4. **Phase 5:** User Acceptance Testing - Complete
5. **Future:** Frontend Development (React)

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